

SECTION A

*Answer **all** questions in the spaces provided.*

1. (a) Draw the **displayed** formula of (*E*)-pent-2-ene. [1]

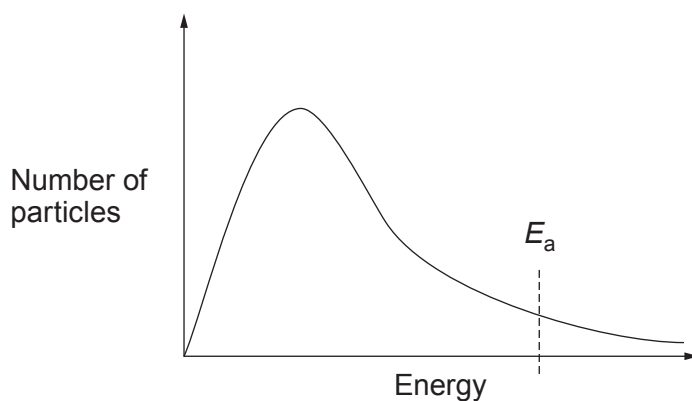
- (b) Draw the **skeletal** formula of 3-chlorohex-1-ene. [1]

2. State the molecular formula of a straight-chained saturated halogenoalkane whose molecules each contain 50 carbon atoms and 2 chlorine atoms. [1]

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3. The diagram shows the distribution of the energies of particles in a gas at a particular temperature.



On the diagram draw a curve to show the distribution of the energies of the same particles at a higher temperature.

Use the diagram to explain the effect of increasing temperature on the rate of a reaction. [2]

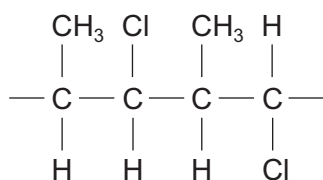
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4. A section of polymer is shown.



Draw the structural formula of the monomer used to make this polymer. [1]



5. Complete the statement below. [2]

To test for the presence of iodine in an organic compound the compound is first heated with This is followed by adding and then to the mixture. If iodine was present in the organic compound then a precipitate is seen.

6. (a) Complete the equation below to show clearly the missing product. [1]



(b) Name the catalyst used in the reaction in part (a). [1]

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